

KUSHAL R V

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EDUCATION

Amrita Vishwa Vidyapeetham — B.Tech Computer & Communication Engineering (CGPA: 7.42) | 2023–Present

PROJECTS

SDR-Based DSSS Signal Detection for Cognitive Radio

- Designed SDR-compatible DSSS generation & detection using GNU Radio and Python.
- Processed complex I/Q signals and modeled RF impairments (AWGN, frequency & phase offset).
- Developed CNN-based detector achieving reliable detection at ~ -30 dB SNR.
- Compared with traditional detectors showing superior robustness.

Sleep vs Awake Classification using Physiological Signals

- Built ML models (RNN, SVM, CNN) for respiratory signal classification.
- Achieved best performance with RNN (AUC 0.92, F1 0.89).

TECHNICAL SKILLS

Wireless & SDR: SDR, Cognitive Radio, DSSS, RF Analysis, I/Q Processing

GNU Radio & DSP: Flowgraph Design, FFT, Baseband Modeling, Channel Simulation

Programming: Python, C, C++, JavaScript

Tools: GNU Radio, NumPy, SciPy, Scikit-learn, Git, Linux

Hardware: ADALM-Pluto SDR, RTL-SDR

CERTIFICATIONS

Oracle OCI AI Foundations | AWS Cloud Foundations | Cisco Networking Basics